

TYPST FOR EFFICIENT TYPESETTING

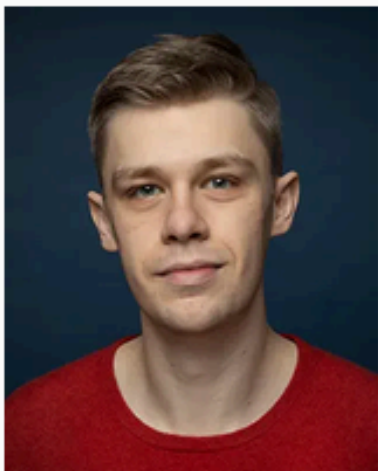
Replacement for LaTeX?

Derek Sollberger

Princeton CSML

Learning Objectives

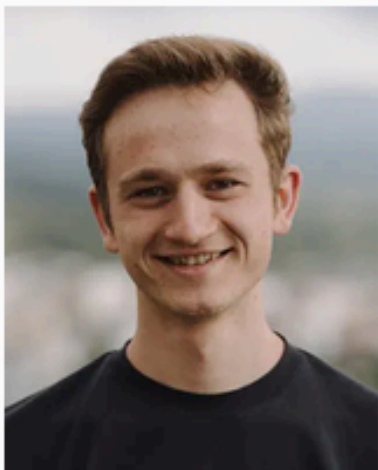
- Introduce Typst for typesetting
- Explore layouts
- Updates from the past year
- Reveal some works in progress



Martin Haug  

FOUNDER, CEO

Martin is a web developer, designer, and the first point of contact for our Enterprise customers. He has a deep background in technologies around desktop publishing and accessibility.



Laurenz Mädje  

FOUNDER, CTO

Laurenz is the main language designer and author of much of the Typst compiler. With a love for consistent and useful systems, changes to our software only make it past him when we can be sure they will be great for our users.

laurmaedje.github.io/posts/comemo/

Laurenz's Blog

What If LaTeX Had Instant Preview?

October 15, 2022

Pitch

- compile in milliseconds (Rust)
- data loading: XML, JSON, CSV
- accessible documents by default
- styling and layouts
- lightweight (about 40 MB)
- secure binary
- actively maintained (about 300 open-souce volunteers)
- integration: CLI, Rust library

Editors

- Typst app (web interface)
- Quarto extensions
- VS Code
- Collabst collaboration

[Start](#)
[Packages](#)
[Templates](#)
[Search](#)
[Browse Categories](#)
[Submit](#)

A plot is worth a thousand words

lilaq

Advanced data visualization

Search 1437 packages and templates

Essential packages

Start quickly with the must-have packages for technical writers

-  **cetz**
Draw figures, tools and charts >
-  **touying**
An object-oriented package for creating presentation slides in Typst. >
-  **unify**
SI units and quantities >
-  **glossarium**
Add a useful glossary >

Draw attention

Your ideas deserve a cool visualization. Add charts, diagrams, and other illustrations

-  **fletcher**
Draw commutative diagrams with nodes and arrows. >
-  **quill**
A library for creating quantum circuit diagrams. >
-  **timelney**
Create Gantt charts in Typst. >
-  **alchemist**
A package to render skeletal formulas using cetz >

Tutorial

Markup

- [blank line for a paragraph break]

`*strong face*`

`_emphasized text_`

`https://nyhackr.org/`

Organization

```
#set heading(numbering: "1.")  
#set heading(numbering: "1.a")
```

= Header 1

== Header 2

=== Header 3

+ ordered

+ list

- unordered

- list

- nested

- list

Math

```


$$\sum_{i=1}^n \sin(\alpha + \beta x)$$


$$\lim_{n \rightarrow \infty} \frac{1}{1+n}$$


$$\sum_{k=0}^n k = 1 + \dots + n \quad \&\& \text{"definition of summation"}$$


$$= \frac{n(n+1)}{2} \quad \&\& \text{"proved by induction"}$$


#let year = 2026
$#year > 42$
$ mat(..#range(1, 17).chunks(4)) $
#math.equation(
  alt: "output equals weights times inputs plus bias",
  block: true,
  $ y = wx + b $,
)
```

Syllabus

Inspiration

Ever since Carlos Scheidegger's talk at **posit::conf(2023)**, I have wanted to try to use **Typst** to create a course syllabus. Over the past day, I have been aided by YouTube videos and the documentation on the Typst website to guide me into creating a fairly creative deliverable. In this blog post, I wanted to elaborate on some design decision and jot down some Typst code just in case I try to teach these materials in the future.

```

#set page("us-letter") //thanks
Europeans!
#set text(
  font: "New Computer Modern",
  size: 14pt
)
#set align(center)
#block(
  fill: rgb("f58025"),
  inset: 14pt,
  width: 100%,
  [= SML 301: Data Intelligence
  = Modern Data Science Methods]
)

```

SML 301: Data Intelligence

Modern Data Science Methods

Instructor: Derek Sollberger

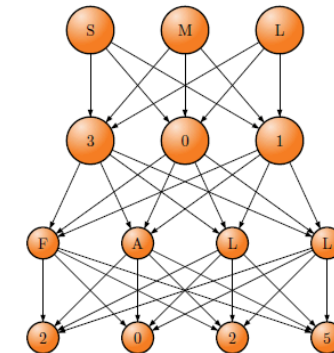
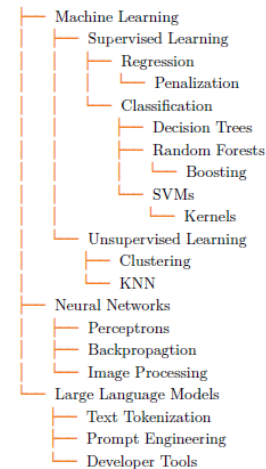
(teacher email)

Post-Docs:

- person name (person email)
- person name (person email)

Teaching Assistants:

- person name (person email)
- person name (person email)
- person name (person email)



```
#import "@preview/treet:1.0.0": *
#text(black, tree-list(
  marker: text(rgb("f58025"))[|— ],
  last-marker: text(rgb("f58025"))
[|— ],
  indent: text(rgb("f58025"))
[|#h(1.5em)],
  empty-indent: h(2em),
)[
  - Machine Learning
    - Supervised Learning
    ...

```

```
#import "@preview/fletcher:0.5.8" as
fletcher: diagram, node, edge
#diagram(
  node-stroke: .1em,
  node-fill:
    gradient.radial(rgb("f58025").lighten(80%
    rgb("f58025"), center: (30%, 20%),
    radius: 60%),
    spacing: 4em,

  node((0.5,1), "S", radius: 1.5em),
  node((1.5,1), "M", radius: 1.5em),
  node((2.5,1), "L", radius: 1.5em),
  edge((0.5,1), (0.5, 2), "-|>"),
```

SML 301: Data Intelligence

Modern Data Science Methods

Instructor: Derek Sollberger

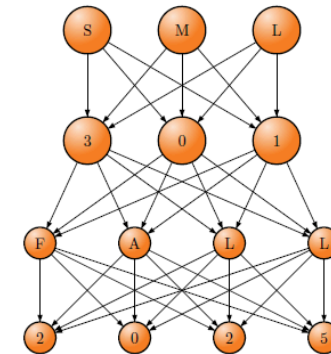
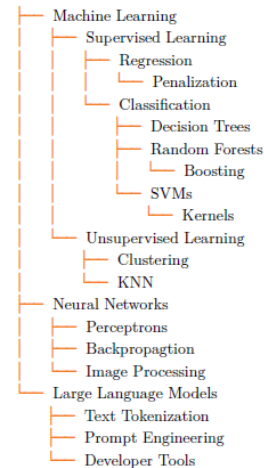
(teacher email)

Post-Docs:

- person name (person email)
- person name (person email)

Teaching Assistants:

- person name (person email)
- person name (person email)
- person name (person email)




```
// Course Description and CLOs
#set align(left)

#grid(
  align: horizon,
  columns: (1fr, 4fr),
  fill: (rgb("f58025"), none),
  gutter: 12pt,
  [],
  [== Course Description
    . . .

    == Course Learning Outcomes
    . . .

    == Prerequisites
    . . .
```

Course Description

This course provides the training for students to be independent in modern data analysis. The course emphasizes the rigorous treatment of data and the programming skills and conceptual understanding required for dealing with modern datasets. The course examines data analysis through the lens of statistics and machine learning methods. Students verify their understanding by working with real datasets. The course also covers supporting topics such as experiment design, ethical data use, best practices for statistical and machine learning methods, reproducible research, writing a quantitative research paper, and presenting research results.

Course Learning Outcomes

1. Understand the main concepts of machine learning in supervised learning, unsupervised learning, and reinforcement learning
2. Deploy and experiment with various machine learning algorithms
3. Critique results and performance of machine learning computations
4. Refine artificial intelligence code and discuss parameter selection
5. Understand the ethical and societal implications of artificial intelligence, including issues such as algorithmic bias, data privacy, and the impact of automation

Prerequisites

1. One semester of data science and statistics
2. Two semesters of calculus
3. Familiarity with R or Python programming is assumed

```
#grid(
  align: center+horizon,
  columns: (1fr, 1fr, 1fr, 1fr,
1fr),
  rows: (1fr, 1fr, 1fr, 1fr, 1fr,
1fr),
  square(fill: rgb("f58025")),
  [_Deep Learning Illustrated_ \ Jon
Krohn et al],
  [_ISLP_ \ James, Witten, Hastie,
Tibshirani, Taylor],
```

Reading List



```
#let lecture_schedule =
csv("SML_301_Fall_2025_lecture_schedule.c
row-type: array).slice(2)
#show table.cell.where(y: 0): set
text(weight: "bold")
#show figure: set block(breakable:
true)
#table(...)
```

Lecture Schedule

Week	Date	Topic	Python	Matplotlib	Precept	Projects
1	3-Sep	introductions			1	
2	8-Sep	linear regression		axes		abstract
2	10-Sep	bias-variance		subplot	2	
3	15-Sep	logistic regression	one-hot	colors		conclusion
3	17-Sep	dimension reduction		aesthetics	3	choose partners
4	22-Sep	decision trees		annotation		methods
4	24-Sep	random forests		title	4	assigned midterm
5	29-Sep	unsupervised learning		arrows		results
5	1-Oct	nonlinear manifolds		layouts	5	midterm code
6	6-Oct	perceptrons	keras			
6	8-Oct	backpropagation	optimizers			midterm due (Oct 10)
7	20-Oct	convolutional neural networks	layers			
7	22-Oct	recurrent neural networks	LSTMs		6	choose project
8	27-Oct	NLP (start)				
8	29-Oct	NLP (word2vec)			7	literature search
9	3-Nov	NLP (ethics)				
9	5-Nov	GANs			8	literature review
10	10-Nov	image processing				
10	12-Nov	topic modeling			9	data glimpse
11	17-Nov	prompt engineering				
11	19-Nov	RAFTs			10	abstract
	24-Nov	Conversation on Industry				
12	1-Dec	Marimo				
12	3-Dec	Gradio				
reading	8-Dec					
reading	10-Dec					

```
#import "@preview/showybox:2.0.4": *  
#showybox(  
  frame: (  
    title-color: rgb("f58025"),  
    body-color: rgb("f58025")  
  ),  
  title-style: (  
    color: black,  
    weight: "bold",  
    align: center  
  ),  
  title: "Typst",  
  [This syllabus document was  
created with Typst!  
  https://typst.app  
],)
```

Typst

This syllabus document was created with Typst! <https://typst.app>

Recent Patches

Accessibility

```
#figure(  
  stack(  
    dir: ltr,  
    spacing: 0.5em,  
    rect[Tagged PDF],  
    text(2em, sym.arrow.long),  
    rect[Accessibility],  
  ),  
  alt: "Diagram with two rectangles.  
The first is labelled 'Tagged PDF'.  
An arrow points to the second,  
labelled 'Accessibility'",  
  caption: [  
    Tags enable PDF accessibility  
  ],  
)
```



Figure 1: Tags enable PDF accessibility

Export & Preview

▼ PDF

Previewed 

Page range

All pages



PDF version ⓘ

Automatic (PDF 1.7)



PDF standard ⓘ

PDF/A-1b

PDF/A-1a

PDF/A-2b

PDF/A-2u

PDF/A-2a

PDF/A-3b

PDF/A-3u

PDF/A-3a

PDF/A-4

PDF/A-4f

PDF/A-4e

 PDF/UA-1

✓ File attachments

Tagged PDF ⓘ



Pretty-print ⓘ



Pretty-printed PDFs consume much more space, but their internal structure is easier to inspect for humans. Primarily useful for developers.

PDFs as Images

```
#figure(
  image(
    "world_cup_semis.pdf",
    alt: "Semi final bracket for
World Cup 2026: France, Spain,
England, Argentina"
  ),
  caption: [
    Semi-final bracket for World Cup
    2026: France, Spain, England,
    Argentina
  ],
)
```



Character-Level Justification

```
#set par(
  justify: true,
  leading: 0.50em,
)
#lorem(256)
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequaleam animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius. Ego autem mirari satis non queo unde hoc sit tam insolens domesticarum rerum fastidium. Non est omnino hic docendi locus; sed ita prorsus existimo, neque eum Torquatum, qui hoc primus cognomen invenerit, aut torquem illum hosti detraxisse, ut aliquam ex eo est consecutus? – Laudem et caritatem, quae sunt vitae sine metu degendae praesidia firmissima. – Filium morte multavit. – Si sine causa, nollem me ab eo delectari, quod ista Platonis, Aristoteli, Theophrasti orationis ornamenta neglexerit. Nam illud quidem physici, credere aliquid esse minimum, quod profecto numquam putavisset, si a Polyaeno, familiari suo, geometrica discere maluisset quam illum etiam ipsum dedocere. Sol Democrito magnus videtur, quippe homini erudito in.

MathML

native support for equations with MathML

- selectable text
- screen readers
- high resolution for browsers

Multiple Bibliographies

- by-*chapter* bibliographies
- by-*topic* bibliographies

Bundle Export

- experimental (beta)

```
#document("index.html", title: [Home])[  
  #title()  
  #link(<blog>)[Go to blog]  
]
```

```
#document("blog.html", title: [Blog])[  
  #title()  
  Welcome to my blog!  
] <blog>
```

Works in Progress

Dwight
Schrute

Creed
Bratton

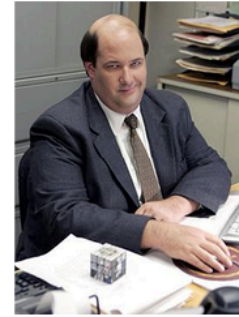
Angela
Kinsey



Meredith
Palmer

Jim
Halpert

Kevin
Malone



Stanley
Hudson

Phyllis
Vance

Pam
Beesley



```
#set enum(numbering: "1).A")
```

+ Which employee is known for his chili?

- + Kevin Malone
- + Creed Bratton
- + Ryan Howard
- + Andy Bernard
- + Darryl Philbin

+ Which employee is known for his "thoughts" blog?

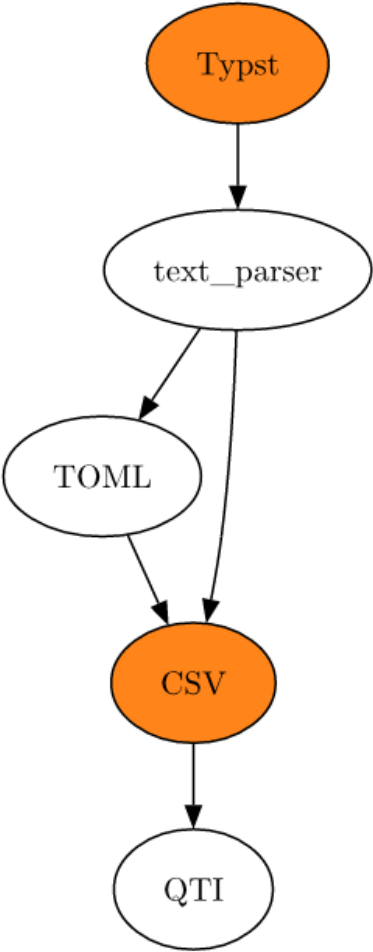
- + Creed Bratton
- + Kevin Malone
- + Ryan Malone
- + Andy Bernard
- + Darryl Philbin

1) Which employee is known for his chili?

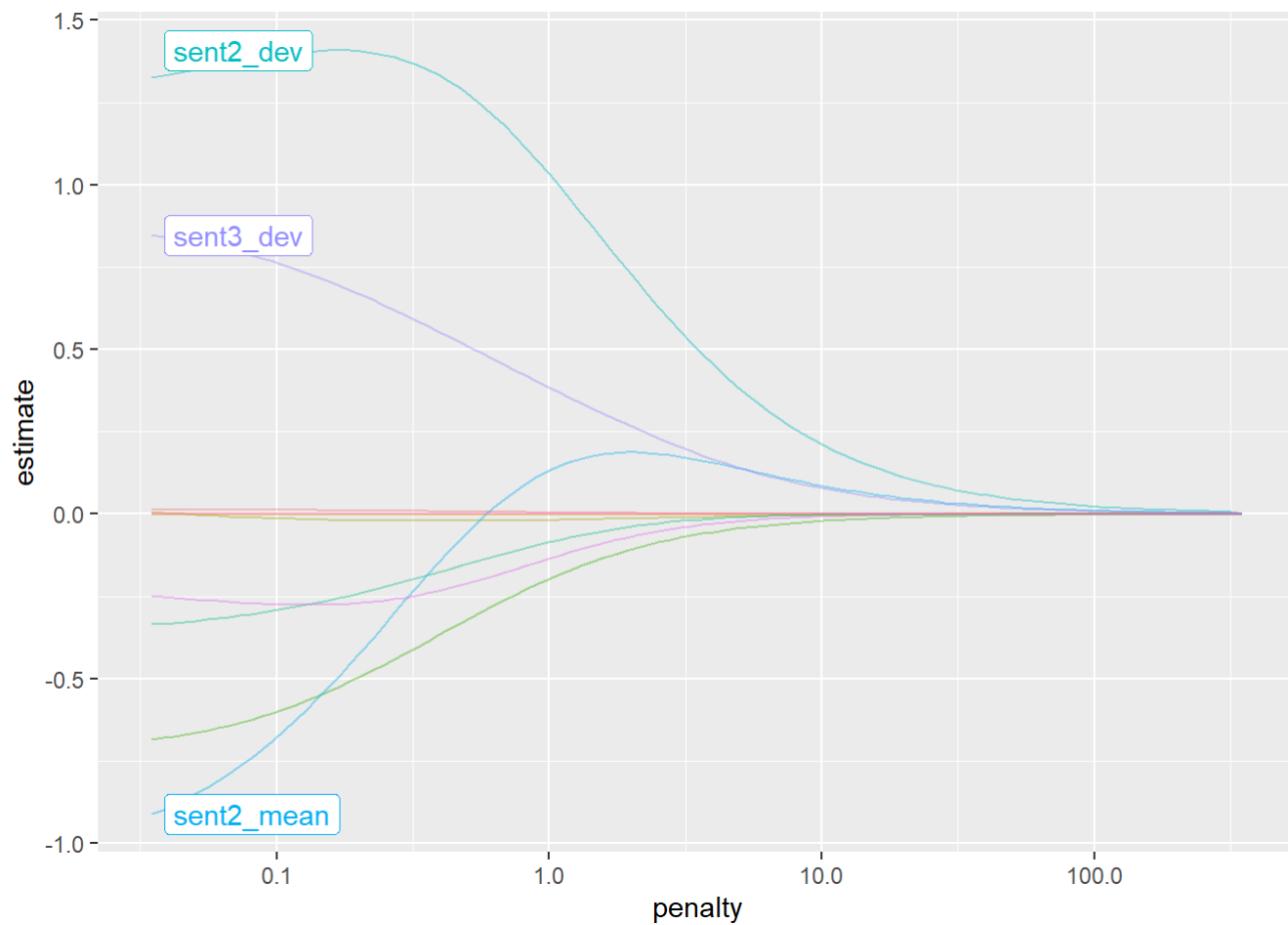
- A) Kevin Malone
- B) Creed Bratton
- C) Ryan Howard
- D) Andy Bernard
- E) Darryl Philbin

2) Which employee is known for his "thoughts" blog?

- A) Creed Bratton
- B) Kevin Malone
- C) Ryan Malone
- D) Andy Bernard
- E) Darryl Philbin

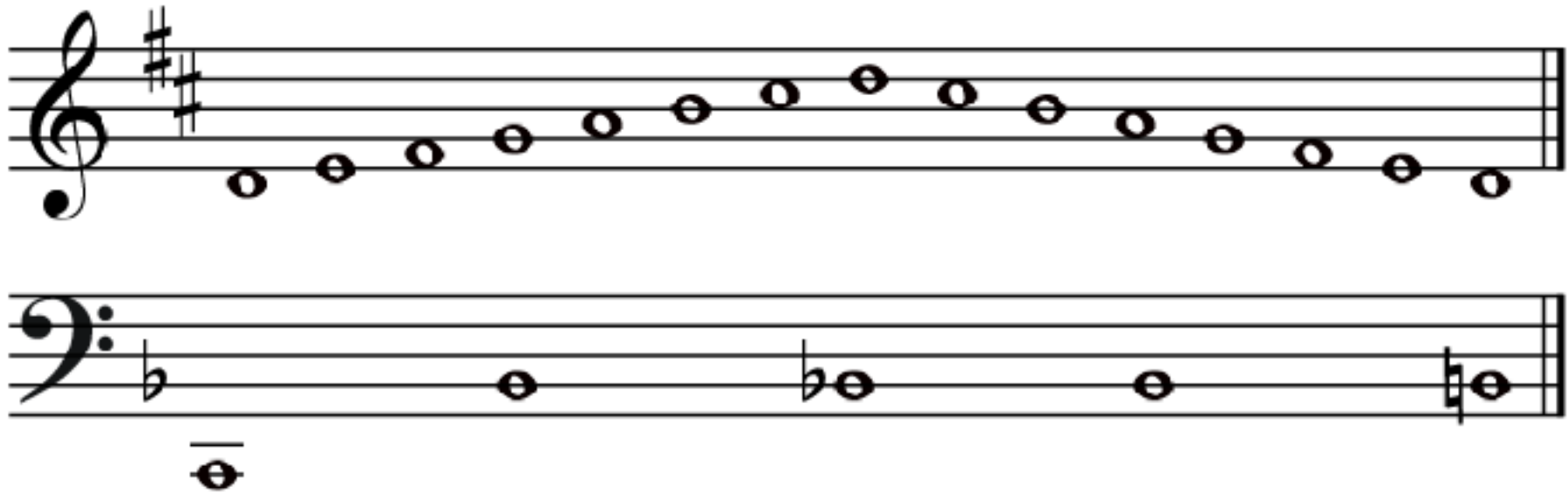


question	correct	alt2	alt3	alt4	alt5
Which employee is known for his chili?	Kevin Malone	Creed Bratton	Ryan Howard	Andy Bernard	Darryl Philbin
Which employee is known for his "thoughts" blog?	Creed Bratton	Kevin Malone	Ryan Howard	Andy Bernard	Darryl Philbin



- $\lambda = 0.1 : y = 7.550 - 0.292X_1 - 0.678X_2 - 0.274X_3$
- $\lambda = 1.0 : y = 7.840 - 0.085X_1 + 0.132X_2 - 0.135X_3$
- **Typst** and *Marimo*?

Staves



Arpeggio 1

```

e|---0---0---0---0---|---0---0---0---0---|
B|---0---0---0---0---|---0---0---0---0---|
G|-----0-----|-----0-----|
D|-----2-----|-----0-----|
A|---3-----3---|---2-----2---|
E|-----|-----|
    
```

Arpeggio 2

```

e|-----0-----0-----0-----0-|-----1-----1-----1-----1-|
B|---1-----1-----1-----1---|---4-----4-----4-----4---|
G|-----0-----|-----0-----|
D|-----2-----|-----0-----|
A|-3-----3---|-2-----2---|
E|-----|-----|
    
```

Finite Simple Group of Order Two

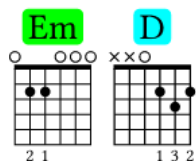
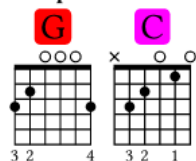
Lyrics by The Klein Four Group

Tab by Music Parsed

Typeset by Derek Sollberger

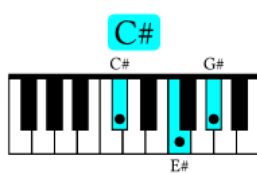
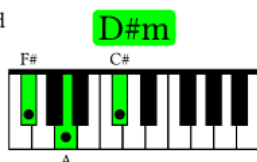
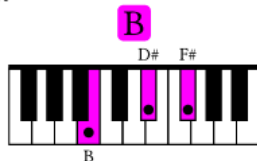
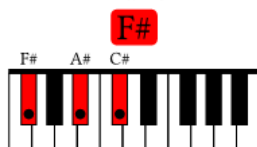
Guitar

Capo: first fret



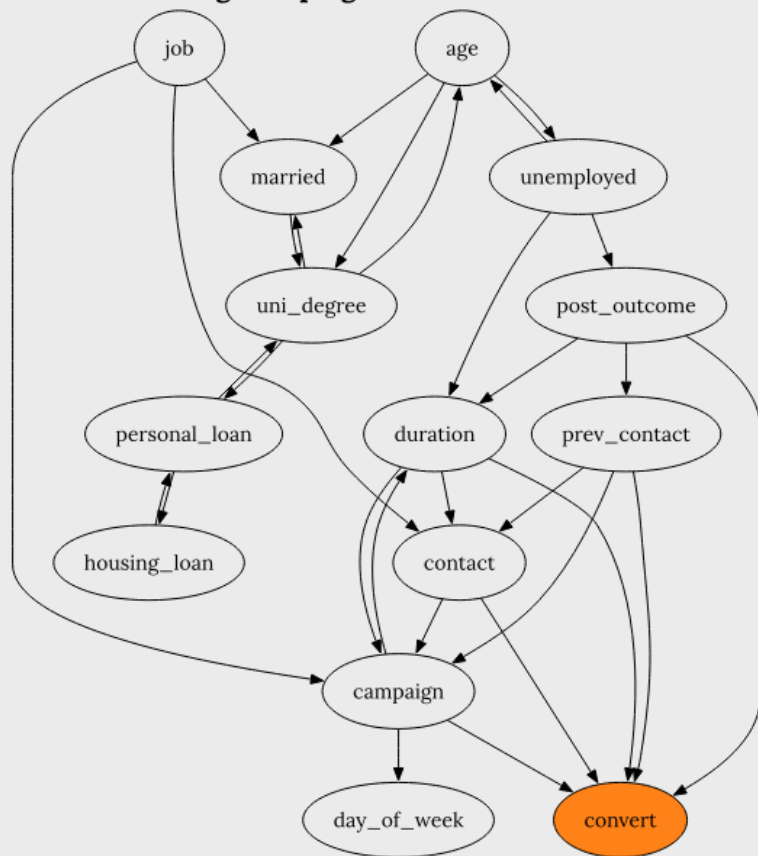
The path of love is never smooth
But mine's continuous for you
You're the upper bound in the chains of my heart
You're my Axiom of Choice, you know it's true
But lately our relation's not so well-defined
And I just can't function without you
I'll prove my proposition and I'm sure you'll find
We're a finite simple group of order two
I'm losing my identity
I'm getting tensor every day
And without loss of generality
I will assume that you feel the same way
Since every time I see you, you just quotient out
The faithful image that I map into
But when we're one-to-one you'll see what I'm about
'Cause we're a finite simple group of order two

Piano



Using the chordx package

Bank Marketing Campaign



- <https://archive.ics.uci.edu/dataset/222/bank+marketing>
- adapted from "Causal Discovery for Hypothesis Generation" by Professor Paul Hunermond

Using the autograph package

In machine learning, **attention** is a method that determines the **importance** of each component in a **sequence** relative to the other components in that **sequence**. In **natural language processing**, **importance** is represented by “soft” weights assigned to each word in a sentence. More generally, **attention** encodes vectors called token embeddings across a fixed-width **sequence** that can range from tens to millions of tokens in size.

—Wikipedia

- color gradients
- variable fonts
- scripting language
- in interactive production?

Donald Knuth on TeX

Of course I do not claim to have found the best solution to every problem. I simply claim that it is a great advantage to have a fixed point as a building block ... I welcome continued research that will lead to alternative systems that can typeset documents better than TeX is able to do. — Donald Knuth, *The Future of TeX and METAFONT* (1990)